Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples

Experiment index

Question and Context

Could a midpoint between 100 and 200 authored ACT examples retain productive accuracy without the same premature-action pressure?

Change

The historical ACT-midpoint candidate reduced the BAYMAX productive quota to 150 while retaining the boundary-training approach. It was evaluated with decision metadata and later strict replay.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT150 / decision metrics	150	46/150 (0.307)	0.520	0.804	0.025	0.052
Historical ACT150 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.050
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052
Historical ACT150 / strict replay	30	6/30 (0.200)	0.467	0.761	0.200	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / decision metrics (150 cases)	0.330	0.000

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / decision metrics (30 cases)	0.343	0.000
Historical ACT150 / strict replay (150 cases)	0.308	0.107
Historical ACT150 / strict replay (30 cases)	0.321	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT150 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT150 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Full-suite success was 46/150 before and after replay; shift success was 6/30. Other metrics changed under strict parsing even though task-success totals stayed the same.

The result did not establish a clean win over ACT-balanced.

Limits

Matching success totals does not imply matching outputs or error types. The surviving historical result is not a complete frozen training manifest. Earlier seed/split defects still apply.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.